

S4I Expression of Interest (EOI) Template for Flagship Scaling Solution Tracks

Section 0 – Title and contact Information

- **Title of the proposed solution track:**
Scaling Private-sector-led Solar-Powered Irrigation Systems (SPIS) for Climate-Resilient Smallholder Farming in Sub-Saharan Africa
- **Title of the AoW2 Flagship that the solution track aligns with:**
Flagship 1.2: Scaling Smart Irrigation and Climate Resilient Water Systems
- **Lead Scientist(s) / Contact Person(s) / Center(s)**
International Water Management Institute
Lead Scientists: Maha Al-Zu'bi, Regional Researcher-Sustainable and Resilient Water Systems (IWMI); Amare Haileslassie, Principal Researcher - Agricultural Water Management (IWMI)
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Contributor Scientists/IWMI:
Seifu Tilahun; Shreya Chakraborty; Deepak Varshney; Muluken Elias Adamseged; Santosh Nepal; Thai Minh; Idil Ires; Tinashe Dirwai.
- **Key CGIAR Center Partners (in the case of multi-Center Eols)**
The International Food Policy Research Institute (IFPRI)
Contributor Scientists / IFPRI:
Bedru Balana, Research Fellow / Natural Resources and Resilience
Oliver Kirui (RF & CPL-Nigeria)
Claudia Ringler (Director, NRR)
- **Countries:**
 - **Scaling activity:** Ethiopia, Kenya, Nigeria
 - **South-South Learning:** India, Bangladesh
- **Technical link to other Science Programs:**
 - CGIAR Science Program on Climate Action
 - CGIAR Science Program on Policy Innovations
 - CGIAR Accelerator on Gender Equality and Inclusion
 - Multifunctional landscape
 - Sustainable Farming
 - CGIAR Digital Innovation

Section 1 – Innovation Package, Value Proposition & Alignment

The Problem to be solved (~ 150 words):

Irrigation is central to climate resilience, yet coverage in sub-Saharan Africa remains extremely low—less than **5% of farmland** is irrigated compared to **40% in South Asia**. Decades of public investment have delivered limited results: in **Ethiopia**, under **5% of 5.3 million hectares** of potential is developed; in **Kenya**, only **222,000 hectares** (16% of potential) are equipped for irrigation, contributing barely **3% to agricultural GDP**; and in **Nigeria**, irrigation covers **less than 1% of cultivated land** (FAO, CGIAR). Public schemes are underperforming, with poor maintenance, weak farmer ownership, and low financial sustainability.

Solar-powered irrigation systems (SPIS) are market-ready, but adoption remains fragmented due to high upfront costs, limited finance, weak supply chains, and inadequate advisory support. A **private-sector-led scaling pathway**, integrating SPIS with **digital advisory tools, blended finance, and inclusive business models**, offers a dynamic solution. Drawing on lessons from **India's PM-KUSUM** (policy design, subsidies, aggregation) and **Bangladesh's IDCOL** (PPP structuring, risk-sharing finance, after-sales networks), this track will catalyze **inclusive, climate-resilient scaling** across Ethiopia, Kenya, and Nigeria.

Scaling ambition (~150 words) for 2030

This Solution Track accelerates the adoption of SPIS to advance Flagship 1.2 outcomes: expanding climate-resilient water systems, reducing fossil fuel dependence, and strengthening smallholder resilience. **By 2030**, through targeting smallholders, cooperatives, women- and youth-led enterprises, and SMEs while leveraging similar initiatives across Ethiopia, Kenya, and Nigeria, it will:

- Enable **1,450,000 smallholders**—with at least 40% women and youth participation—to access solar irrigation pumps, tailored climate/blended finance packages, bundled innovations (digital water advisory, input optimization, and climate services), and strengthened market linkages that improve productivity, incomes, and resilience.
- Make **900,000 solar irrigation units** more accessible and affordable to SHFs via inclusive PPPs, inclusive finance, and market linkages tailored to local demand.
- Catalyze **USD 420,000,000 (0.42 billion USD)** in blended finance and SME investment, reducing market and financial access gaps.
- Strengthen extension capacity: **9,000 extension** workers trained.
- Increase high-value crop production across **850,000 ha** and reduce soil- and water-limited yield gaps by 30–40%, depending on context.
- Reduce household food insecurity and increase dietary diversity (nutrition) **for 4,450,000 people**.
- Create **150,000 jobs** for youth and women.

Country-specific bundles address local needs:

1. **Ethiopia:** Solar-powered pumps + PAYOWN+ credit guarantee for de-risking private and financial institutions + digital pre and post-sale service + solar suitability mapping
2. **Kenya:** Solar-powered pumps + PAYOWN/PAYGO+ solar suitability mapping + credit guarantee for de-risking private + carbon and blended finance.
3. **Nigeria:** Solar-powered pumps + PAYOWN+ solar suitability mapping+ result-based and targeted financing + private sector innovation challenge grants + smart marketing strategies.

South-south learning and capacity sharing:

1. **India:** Policy clinics + subsidy targeting models (PM-KUSUM) + bulk procurement/aggregation mechanisms + private-sector partnership exchanges
2. **Bangladesh:** PPP structuring (IDCOL) + risk-sharing/blended finance design + vendor/after-sales service models + women-inclusive farmer participation

Innovation & Value Proposition (~150 words)

The Solution Track offers scalable, context-specific bundles of SPIS combined with financial and institutional innovations. **IWMI** brings global leadership in solar suitability mapping, water resource assessment, and agribusiness technical assistance, coupled with experience in market demand segmentation, value chain stakeholder engagement, and co-design of inclusive business models such as PPP and PAYGo/PAYOwn. These models unlock farmer demand, mobilize private investment, and ensure last-mile access to farmer communities, including women and youth. **IFPRI** complements this with robust economic and policy analysis, developing decision-support tools that integrate solar suitability with profitability metrics linked to cropping patterns, production costs, and market access. IFPRI also contributes to policy engagement, market-based instruments, and enabling conditions for efficient irrigation supply chains.

CGIAR's proven partnerships demonstrate scalability: India's PM-KUSUM program (>800,000 solar pumps), Bangladesh IDCOL PPP models (>1,500 pumps), off-grid solar and blended finance in Kenya and Ethiopia with SDC (SolAR-II), and co-design scaling solar-powered irrigation in Nigeria with the Bank of Agriculture and the World Bank-supported ACREsAL project, and with the Ministry of Agriculture and Food Security and Kano and Oyo state governments to capitalize on result-based financing.

Together, IWMI and IFPRI draw on CGIAR's proven partnerships: PM-KUSUM in India (>800,000 pumps), IDCOL in Bangladesh (>1,500 pumps in a fee-for-service model), SolAR-II in Kenya/Ethiopia, and co-design with Nigeria's Bank of Agriculture, ACREsAL, and state governments. This integrated approach ensures inclusive, cost-effective, and sustainable scaling of SPIS across Africa leveraging lessons learned from South Asia.

Customers and alignment (~100 words)

Institutional customers include ministries of agriculture, water, and energy in Kenya, Ethiopia, and Nigeria, which seek evidence-based policies, innovative financing models, and technical

solutions aligned with national irrigation strategies (e.g., Kenya’s NISIP, Ethiopia’s 10-year solar irrigation plan, and Nigeria’s ACREsAL and DARES initiatives). **Development partners** such as the World Bank, AfDB, EU, FCDO, and SDC, together with regional and local financial institutions, benefit from validated models, market intelligence, and investment-ready opportunities. **Private-sector partners**—including SunCulture (Kenya), Pumptech (Ethiopia), agribusinesses, distributors, and local banks—gain expanded markets and service contracts, with additional partners to be identified and engaged. **End-users**—farmers, SMEs, and cooperatives—access tailored SPIS bundles, with emphasis on women and youth.

Evidence of Impact (~150 words)

Evidence from CGIAR and partners shows that solar-powered irrigation can be scaled sustainably and profitably across Africa and South Asia. In **Nigeria**, an IFPRI–IWMI study identified **5.04 million hectares of profitable groundwater-fed irrigation potential, mostly viable with SPIS**, while bilateral projects applied **Human-Centered Design** to co-create solutions with government, finance, and private-sector actors. IWMI pilots and IFPRI policy analysis informed frameworks for scaling vegetable value chains, complementing national programs such as the World Bank–backed \$750 million **DARES facility**. In **Kenya**, partnerships with SunCulture and other private actors reached thousands of smallholders by linking pumps with finance and advisory services. In **Ethiopia**, pilots with the Agricultural Transformation Institute validated **PAYOWN financing, credit guarantees, and digital tools**, while IFPRI **advanced gender-responsive irrigation analysis and groundwater governance innovations**.

Beyond Africa, the SDC–SoLAR program has generated evidence on groundwater sustainability in South Asia, with peer-reviewed studies confirming SPIS adoption readiness, affordability, and inclusive impacts (Burney et al., 2010; Xie et al., 2021; Sarjer et al., 2025). IWMI has also provided direct advisory support in India and Bangladesh: in India, IWMI’s pilots in Gujarat informed the design of the national grid-connected solar irrigation program under PM-KUSUM, and in Bangladesh, IWMI’s pilots enabled IDCOL to leverage KfW funding to scale grid-connected SPIS. Together, this evidence demonstrates that SPIS can scale successfully, but challenges of affordability, equitable access, sustainable groundwater use, and coordinated policy and market support must be addressed for inclusive, cost-effective, and climate-resilient outcomes.

Section 2 – Scaling pathway, sustainability, partnerships & risks

Scaling Pathway and Sustainability (~250 words)

Private Sector & Community Partnerships for Inclusive Solar-powered Irrigation Scaling

Advances a private-sector-led scaling model anchored in strong community partnerships to ensure inclusivity and long-term sustainability. By shifting reliance away from public subsidy-driven schemes, it mobilizes **private capital, SME innovation, and blended finance** to expand access to affordable SPIS technologies. **Communities, cooperatives, and water user associations** co-manage PAYGo/PAYOwn systems, enhancing ownership, accountability, and system longevity. This model embeds local supply chains for distribution, repair, and after-sales

service—overcoming chronic challenges of equipment availability and maintenance—while **digital platforms** deliver tailored irrigation scheduling, crop advice, and climate adaptation support to smallholders.

IWMI provides leadership on market and service model design, solar suitability mapping, and development of **community-tailored irrigation bundles**. It also supports policy reform, enabling blended finance instruments, and strengthens governance frameworks for groundwater regulation and water–energy–food nexus risk management. IFPRI enhances this with **socio-economic and market assessments, gender and equity analysis, and policy engagement** to shape enabling environments for inclusive financing solutions.

Sustainability is embedded through: **(i) co-financing mechanisms** (lease-to-own, credit guarantees, carbon and results-based finance) that reduce risks for farmers and investors; **(ii) local institutional strengthening**, linking cooperatives and SMEs into national value chains; and **(iii) policy integration** that embeds SPIS into national climate, energy, and agricultural strategies.

Risks—such as inequitable access, over-extraction of groundwater, or weak private-sector participation—are proactively mitigated through inclusive business models, groundwater governance frameworks, and blended finance tools that de-risk investment. This coordinated approach ensures SPIS scaling is not only rapid but also inclusive, cost-effective, and climate resilient.

Partnerships & Leverage (~300 words)

This Solution Track is anchored in CGIAR’s convening power and proven ability to bridge research, policy, and markets. **IWMI** leads on scaling solar irrigation innovations with private institutions, financing models, and inclusive and enabling policies, backed by tools such as **solar suitability mapping, pump-sizing applications, and service delivery dashboards** that guide investment and implementation. **IFPRI** strengthens this with **socio-economic and market assessments, equity and gender analysis, and policy engagement** to ensure financing solutions are inclusive, profitable, and sustainable. Together, they provide the technical depth and policy reach needed to scale SPIS equitably across Africa.

Externally, the Track is underpinned by **strong national partnerships** that demonstrate both demand and readiness. In **Kenya**, collaboration with the State Department for Irrigation, SunCulture, and others to develop a blended finance facility supports the National Irrigation Sector Investment Plan (NISIP) and an enabling environment through inclusive tariff structures. In **Ethiopia**, partnerships with the Ministries of Agriculture, Irrigation and Lowlands, Water and Energy, Finance, and Planning and Development, alongside the 10-year solar irrigation and PPP strategies, position the Track within national planning frameworks. In **Nigeria**, alliances with the Federal Ministries of Water Resources and Agriculture, the Bank of Agriculture, Kano and Oyo state governments, and the World Bank–backed DARES and ACRoSAL projects create a powerful platform for scaling.

Bilateral projects extend this leverage: SoLAR-II (SDC), IFAD, and World Bank PPP pilots in Ethiopia and Kenya validate PAYGo/PAYOwn financing, cost-sharing, and service bundling. IWMI's work through the WE4F Hub in Southern and Central Africa, and IFPRI's projects on groundwater governance, nutrition, gender, and irrigation economics in Ethiopia, Nigeria, and Kenya, strengthen the evidence base for scaling.

The Track institutionalizes a South–South learning framework, drawing on India's PM-KUSUM and Bangladesh's IDCOL models, while convening governments, financiers, and private companies for peer-to-peer exchange. Private-sector partners—including SunCulture, Pumpteck, Green Scene, Tech2, Lupiya, and Innovative Engineering Services—bring distribution, finance, and after-sales networks. Donors and IFIs—World Bank, AfDB, EU, SDC, FCDO, MFAT, GF—represent more than **USD 2.5 billion** in ongoing investments ready to be leveraged.

Risks and Bottlenecks (~250 words).

Scaling solar-powered irrigation bundles through the **Private Sector & Community Partnership** pathway faces financial, policy, operational, social, and environmental risks across target countries.

Financial: High upfront costs, limited access to credit, and limited liquidity among smallholders and SMEs. Mitigation includes PAYGo/PAYOwn models, blended finance with donor guarantees, targeted subsidies, and partnerships with MFIs and private suppliers.

Policy & Institutional: Slow implementation of guidelines, lack of solar pumps standards, inconsistency of policies, and fragmented and overlapping of responsibilities across ministries create uncertainty. Mitigation involves multi-stakeholder PPP steering committees, policy dialogue, subsidy realignment toward clean irrigation and national strategies.

Operational: Weak distribution networks limited pre- and post-sales service, and technician shortages reduce adoption and trust. Mitigation includes accrediting vendors, knowledge and capacity development, training programs, supporting SMEs with working capital, and establishing regional hubs.

Social: Risks include exclusion of women, youth, and marginalized farmers or elite capture. Mitigation prioritizes pro-poor targeting, quotas, group-based lending, and inclusive extension/advisory services.

Environmental: Groundwater overuse and inefficient irrigation threaten sustainability. Mitigation integrates water-use permits into PPPs, promotes efficient irrigation (drip/fertigation), and links digital advisory with solar suitability mapping, solar pump sizing.

South–South learning accelerates policy, regulatory, and service capacity by transferring proven models across contexts. Together, these measures reduce adoption risk, enhance inclusion and sustainability, and strengthen farmer and investor confidence, directly supporting this scaling pathway.

Section 3 – Responsible scaling trajectory & achievable scaling target

Readiness & Responsible Scaling (~150 words)

Solar-powered irrigation (core innovation) is scaling-ready, having progressed beyond prototypes and validated under real-world conditions with private-sector partners (Hauke & Magombeyi, 2022; IWMI pilot data, 2023–25) (See Annex section for IPSR reports). Adoption drivers—affordability, reliability, and policy alignment—are proven, with primary bottlenecks in finance, policy, and institutional capacity rather than technology.

Complementary innovations (PAYOWN+ credit guarantee for de-risking private and financial institutions + digital pre- and post-sale service + solar suitability mapping, PAYOWN/PAYGO + carbon and blended finance, result-based financing, smart marketing strategies).

Under Pathway 1 (Private Sector & Community Partnerships), innovations are bundled into community-managed service models with digital advisory support, inclusive financing, and strengthened supply chains.

Responsible scaling includes inclusivity (financing, training, advisory for women and youth), environmental safeguards (WA+, efficient irrigation, solar suitability mapping), and policy integration, preparing innovations for socially inclusive, environmentally sustainable, and mainstream adoption across national and regional programs.

Achievable Target, Resources & Activities (~250 words)

By 2030, this Solution Track will reach 1,450,000 smallholders, with at least 40% women and youth participation, by deploying 900,000 solar irrigation units through inclusive PPP models tailored to local demand. It will mobilize USD 420 million in blended finance and SME investment, train 9,000 extension agents, increase high-value crop production across 850,000 hectares, and reduce yield gaps by 30–40%. Policy uptake in at least four national programs, reduced food insecurity and improved dietary diversity for 4.45 million people, and 150,000 new jobs for women and youth will underscore the impact.

Resources and capabilities underpinning delivery include: CGIAR scientists (IWMI, IFPRI), national partners in finance, gender, and inclusion; IWMI’s scaling hubs in Ethiopia, Nigeria, and Kenya; and catalytic S4I financing to unlock donor and IFI co-investment. Technical assets—solar pump-sizing tools, suitability maps, water accounting, digital advisory platforms, and irrigation service dashboards—provide data-driven scaling guidance.

Activities:

- **Training & Advisory:** Build capacity of extension staff, cooperatives, and farmer groups in SPIS operation, groundwater stewardship, business skills, and climate-smart agronomy; scale digital advisory platforms.
- **Supply Chain & SME Support:** Incubation and technical assistance for SMEs to strengthen pump distribution, repair, and after-sales services; local entrepreneurs incentivized to build rural service networks.

- **Finance & Market Models:** Development of PAYGo/PAYOwn packages, credit guarantees, and carbon/result-based finance; IFPRI develops tools to align financing with farm profitability, input costs, and crop choices.
- **Policy Engagement:** Multi-stakeholder policy clinics, co-creation of irrigation strategies, and integration of SPIS into agricultural, energy, and climate action plans.
- **Capacity Development & South–South Exchange:** Structured exchanges with India and Bangladesh on subsidy design, PPP structuring, and service networks; toolkits, case studies, and communities of practice disseminated.
- **Monitoring, Learning & Adaptation:** Rigorous tracking of adoption, gender/youth participation, profitability, and environmental outcomes, with lessons reinvested into scaling models.

Timeline, Milestones & Monitoring (~200 words)

Year	Milestones	Monitoring & Learning
2026	Reaching out to 150,000 farmers; Deploy 100,000 solar irrigation pumps; Establish blended finance facilities; MoUs with governments and private-sector partners.	Adoption tracked via farmer surveys, IPSR framework, and supplier data (IWMI and IFPRI); gender and youth participation monitored (IWMI), Link to AoW3-policy uptake progress tracked (IWMI and IFPRI); assess service reliability and advisory adoption (IWMI and IFPRI); and process-based outcome assessment to ensure adaptive management along scaling processes
2027	Reach 300,000 farmers; mobilize \$100 M co-financing; extension capacity for 1,000 workers and agencies trained;	Environmental monitoring via WA+ and solar suitability maps (IWMI); policy implementation and financing mechanisms evaluated (IFPRI); lessons captured for adaptive scaling across activities (IWMI and IFPRI).
2028	Reach 350,000 farmers; mainstream solar irrigation in ≥2 national policies/programs; leverage \$150M+ investments; extension capacity for 2,000 workers and agencies trained, Reduce household food insecurity and increase dietary diversity (nutrition) for 1 million people , document and share global lessons.	Continuous MELIA-based monitoring; assess jobs creation, GHG reductions (IWMI/IFPRI); adaptive learning shared regionally via EE Labs; pathway-specific insights captured from government-led and private/community-led models; nutrition and diversification outcomes monitored (IFPRI); Evidence on solar-powered irrigation scaling impact in Sub-Saharan Africa and South Asia documented and shared for the south-south learning

Monitoring & Learning:

- **Adoption:** Farmer surveys, IPSR framework, supplier and extension data (IWMI).
- **Environment:** WA+ water accounting, solar suitability mapping, efficient irrigation adoption.
- **Gender & Youth:** Disaggregated adoption metrics (IFPRI).
- **Policy & Finance:** National policy integration, tariff/market reforms, finance uptake (IWMI).
- **Monitoring and Learning:** Lessons synthesized through EE Labs and South–South exchanges to inform adaptive scaling, pathway optimization, and policy integration.

Annexes

- IPSR Innovation Profile (if available):
 - Sustainable Finance for Off-Grid Solar Irrigation: IPSR Innovation Profile
 - Solar Irrigation Pump (SIP) Sizing Tool: IPSR Innovation Profile
- Scaling Readiness assessment results
 - <https://www.spotlightnepal.com/2023/07/03/iwmi-organized-workshop-solar-pump-sizing-tool-scaling-nepal/> (Workshop report available)
 - CGIAR Results Dashboard - CGIAR innovation readiness for solar irrigation in Ethiopia (7)
 - Solar-based irrigation bundle profile and scaling in Ghana.
<https://hdl.handle.net/10568/128163>
 - Irrigation for smallholder farmers - CGIAR scaling maturity level; solar; irrigation farmers led irrigation
- Scaling Scan results (if available)
 - Prior project reports IFPRI-led assessment of the potential of FLID in the ACRSAI Project. Unlocking the potential of farmer-led irrigation development in central and northern Nigeria: What ... IFPRI blog: Overcoming policy, finance, and technology supply constraints to unleash small-scale irrigation in ... Lee, Yeyoung; Bryan, Elizabeth; Mason, Nicole M.; Hassen, Ibrahim Worku; Theriault, Veronique; and Ringler, Claudia. 2025. Does small-scale irrigation affect women's time allocation? Insights from Ethiopia. World Development 196 (December 2025): 107106. <https://doi.org/10.1016/j.worlddev.2025.107106>
 - Ringler, Claudia; McCartney, Matthew P.; and Hafeez, Mohsin. 2025. What do we know about the future of water in food systems? In What do we know about the future of food systems? eds. Keith Wiebe and Elisabetta Gotor. Chapter 10, Pp. 54-59. Washington, DC: International Food Policy Research Institute. <https://hdl.handle.net/10568/175232>

- Media or other evidence of prior scaling work that can be accelerated
 - [IWMI supports climate-smart agriculture in Ethiopia - IWMI](#)
 - [Scaling solar-based irrigation bundles in Ethiopia: a market linkage pathway](#)
 - [Solarizing will make agriculture resilient in South Asia and East Africa - IWMI](#)
 - Mapping the suitability of solar energy-based irrigation pumps in Ethiopia
 - [Does solar irrigation threaten groundwater sustainability? Evidence from India and Bangladesh](#)
 - [Solar Irrigation for Agricultural Resilience \(SoLAR\) in South Asia: impact pathways](#)
 - [Mapping tool helps private sector identify high-potential locations for solar irrigation pumps | Water, Land and Ecosystems](#)
 - [Business model scenarios and suitability: Smallholder solar pump-based irrigation in Ethiopia. - IWMI](#)
 - [IWMI and climate tech startup SunCulture establish landmark partnership to advance sustainable solar irrigation in Kenya - IWMI](#)
 - [SunCulture and IWMI Partner to Drive Sustainable Water Management in Solar Irrigation - SunCulture](#)
 - [Solarizing will make agriculture resilient in South Asia and East Africa - IWMI](#)
 - [IWMI and climate tech startup SunCulture establish landmark partnership to advance sustainable solar irrigation in Kenya - IWMI](#)
- MELIA studies, impact assessments, reports, publications
 - [Unlocking the potential of farmer-led irrigation development in central and northern Nigeria: What does it take?](#)
- List of Bilateral / Country:

IWMI Bilateral	
Bangladesh	Solar Energy for Agricultural Resilience (SoLAR) Phase 2
India	Solar Energy for Agricultural Resilience (SoLAR) Phase 2 SDC : Solar Energy for Agricultural Resilience (SoLAR) Phase 2 builds upon learnings and experiences from Phase 1 of SoLAR (2019-2024) in South Asia while expanding its scope to East Africa through meaningful south-south collaborations, USD 5.75 million GIZ : Solar Energy for Rural Livelihood Programs (SE4RL), The overall goal of the project is to catalyze and support improved decision making in the process of scaling and mainstreaming solarization of agriculture in India, specifically through the Government of India's PM-KUSUM initiative, and Nepal Terai. IKI : IWMI, as a consortium partner, will be working with GIZ for the design and implementation of a 20m EUR project focused on the promotion of photovoltaic expansion on agricultural land in India
Ethiopia	Solar Energy for Agricultural Resilience (SoLAR) Phase 2 SDC solar pump

	World bank financed PPP involvement in irrigation service provision, IFAD, SDC, Gate Foundation MFAT , Nature based solution for sustainable and inclusive development (NSSID)
Kenya	Solar Energy for Agricultural Resilience (SoLAR) Phase 2
Nigeria	TAAT III by AfDB (2.5 million USD) B- ACREsAL by WB (1 million USD)

IFPRI Bilateral	
Nigeria	TAAT III, WB- ACREsAL, & Gates Foundation supported Suitability maps (\$1 million)
Ethiopia	BHA PSNP (public works on watershed & SSI), Ground water governance, & Integrative SSI maps, Groundwater policy work
Kenya	Economy-wide assessment of groundwater development; climate finance for food system transformation (World Bank)

- Partner commitments (MOUs, LOIs, letters of support)
Available and can be provided, below examples from Nigeria and India.

Example from Nigeria

Meeting summary

Exploring collaboration between IWMI and the Bank of Agriculture on scaling solar-powered irrigation for smallholder farmers in Nigeria

Virtual Meeting (Zoom): August 8, 2025

Participants

- **IWMI:** Olufunke Cofie (IWMI Africa Director), Adebayo Oke, Christopher Agyekumhene, Kekeli Gbodji, Seifu Tilahun
- **BOA:** Ayodeji Sotinrin (MD, BOA), Renee Phil-Agbasi, Deborah Owhin,

Background

The International Water Management Institute (IWMI) is advancing efforts to scale solar-powered small-scale irrigation systems (SPIS) globally, including in Nigeria. SPIS has proven effective in boosting smallholder farmer productivity, resilience, and incomes across the Global South. In Nigeria, this work is being pursued through **CGIAR Scaling for Impact (S4I) program**. Within this context, the Bank of Agriculture (BOA) expressed interest in exploring partnerships with IWMI. BOA is a Federal Government-owned development finance institution with over 50 years of experience supporting agriculture and rural development. With a strong mandate to empower rural communities, BOA works with partners who share its vision of delivering innovative financial solutions that transform livelihoods nationwide.

Meeting highlights

The meeting explored opportunities for collaboration between IWMI and BOA, focusing on three strategic areas:

1. **Partnership development** – Co-designing and scaling SPIS in Northern Nigeria through a strategic collaboration.
2. **Financing and policy innovation** – Unlocking financing models and policy pathways to accelerate SPIS adoption, drawing on lessons from other countries.
3. **Product expansion** – Supporting the expansion of BOA's financial products to strengthen farmer-led irrigation development across Nigeria.

Presentation highlights

Olufunke Cofie, IWMI Africa Director for Research Impact, and Seifu Tilahun shared IWMI's experiences in farmer-led irrigation, emphasizing the potential for solar-powered irrigation in Nigeria. They identified suitable deployment areas, highlighted existing government initiatives, and drew lessons from Asia and Africa, especially Ghana's cocoa sector, where IWMI partnered with Access Bank to finance solar irrigation systems.

Discussion highlights

- **Partnership commitment:** Ayodeji Sotinrin, Managing Director of BOA, expressed strong interest in partnering with IWMI to scale solar irrigation in Northern Nigeria. His focus areas included increasing farmer access and ownership of solar irrigation solutions, creating jobs for youth and women, and improving rural livelihoods.
- **Ambition and investment:** BOA aims to provide irrigation kits to **500,000** farmers and mobilize up to **USD 300 million**, targeting a **20-million-ton increase** in food production.
- **Holistic approach:** Participants emphasized the need for integrated solutions addressing governance, maintenance, and sustainable irrigation practices.
- **Program development:** Agreement to develop a sustainable program, beginning with terms of reference, and activities on appropriate irrigation kits and hotspot areas for implementation. Participants also discussed the importance of blended financing and leveraging various funding instruments to raise the necessary funds for the intended Programme. IWMI and BOA will jointly address barriers to technology and finance access for smallholder farmers.

BOA requests from IWMI

1. Technical specifications and reliable brands for solar irrigation equipment.
2. Guidance on sustainable scaling pathways.
3. Capacity building for farmers and BOA staff.
4. Support in establishing maintenance and management systems for irrigation kits.

Agreed action points

Coordination: Renee Phil-Agbasi and Deborah Owhin will serve as BOA's main contacts, while Adebayo and Seifu are the main IWMI contacts.

Joint technical team: IWMI and BOA will establish a working team to co-design the program on SPIS

Targeting: Identify farmer groups and hotspot areas for implementation.

Financing: Develop a blended financing model leveraging multiple instruments.

Partnerships: Explore collaboration with additional financial institutions.

Data & impact tracking: Build a shared dashboard for monitoring data, impacts, and stakeholder engagement.

Next steps

- Develop **terms of reference** and sign a **formal partnership agreement**.
- Develop and implement a collaboration project on suitable irrigation kits hotspots areas.
- Co-develop a program leveraging IWMI's global experience to ensure governance, maintenance, and sustainability.
- Share data on Nigeria's solar irrigation potential to guide BOA's financing strategies.
- Strengthen evidence generation on cost-effective and reliable solar irrigation solutions.
- Provide guidance on water governance systems for sustainable small-scale irrigation in Nigeria.

Contact Persons: **IWMI:** Adebayo Oke, **BOA:** Renee Phil-Agbasi,

Example from India

No. 4548 /Bio General/2024-25
Uttar Pradesh New and Renewable Energy Development Agency,
Vibhuti Khand, Gomti Nagar, Lucknow.

Date- 12th November, 2024

To

Dr. Deepak Varshney,
International Water Management Institute (IWMI), New Delhi

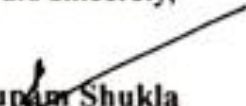
Subject: Approval for advisory services in PM KUSUM, Pilot Project and other RE initiatives

Sir,

With reference to the subject mentioned above, and in response to your email dated 11th November 2024, we hereby acknowledge and provide you the consent for advisory services which has no financial and legal obligations to UPNEDA. Also, this will help for the pilot project focused on transitioning diesel-powered boats to renewable energy (RE) boats, as well as the implementation of the PM-KUSUM scheme.

We believe that your expertise in these programs will greatly benefit UPNEDA, helping us advance our efforts in the aforementioned projects, along with other renewable energy initiatives. We look forward to collaborating with you and leveraging your guidance for the successful execution of these endeavours.

Yours sincerely,


Anupam Shukla
(Director, UPNEDA)